

DANIEL LIAN

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🐙 github.com/GlutenFreeGrapes

Education

University of Michigan

Master of Science in Engineering in Computer Science & Engineering

Aug. 2026 – May 2027

Ann Arbor, Michigan

University of Michigan

Bachelor of Science in Engineering in Computer Science - GPA: 3.7/4.0

Aug. 2023 – May 2026

Ann Arbor, Michigan

Relevant Coursework

- Cryptography
- Foundations of Computer Science
- Discrete Math
- Data Structures
- Ethics and AI
- Artificial Intelligence
- Conversational Artificial Intelligence
- Computer Vision
- Computer Security
- Software Design
- Data Science
- Operating Systems
- Computer Organization
- Computational Linguistics
- Problem Solving

Experience

Fairfax Collegiate

Instructor

June 2023 – August 2025

McLean, VA

- Instructed various STEM classes to students over 15 weeks, teaching student of various ages robotics, basic programming skills, advanced math concepts including middle school math and algebra, fundamentals of game design.
- Honed communication skills by mentoring junior staff, created friendlier team-based learning environment to adapt to students' diverse learning styles.

Projects

Machine Learning Mail Sorter | *Python, Selenium, Pandas, HTML, Flask, Google OAuth/API* **June 2025 - August 2025**

- Designed system for and developed an email classifier using Google OAuth login capable of categorizing 4000 emails per minute with automatic mail organization and spam detection.
- Written using Python backend with Naive Bayes classifier, uses GMail API to collect and sort emails, wrapped in Flask app using a simple and easy-to-follow HTML interface.

Cities Chain Discord Bot | *Python, Pandas, MariaDB, SQL*

July 2023 - Present

- Designed and implemented a scalable Discord game supporting thousands of players across 700+ Discord servers using efficient SQL backend data management in MariaDB for seamless gameplay.
- Applied Agile software maintenance principles and used player feedback to ensure robust gameplay experience.

Election Prediction Algorithm | *Python, Jupyter Notebook, TensorFlow, Keras, Pandas, NumPy* **August 2022 - May 2023**

- Implemented Python Tensorflow model in Jupyter Notebook to predict 85% of 2022 United States congressional winners using weighted average of 538 polls, demonstrating robust predictive analytics capabilities.
- Presented research and findings to audience of 40 at tjSTAR technology research symposium.

University of Pennsylvania Wharton Data Science Academy | *R*

July 2022

- Trained neural networks on large text dataset to determine star ratings of Yelp reviews in R, gaining practical experience in Natural Language Processing (NLP) and model interpretation.
- Built and analyzed datasets with R to assess company market caps and stock performance relative to the S&P 500.
- Collaborated with a team to develop a predictive model for real-world stock price changes from insider trading.

Technical Skills

Languages: Python (Pytorch, TensorFlow, Selenium, Flask, Pandas, NumPy, scikit-learn), C/C++, Java, SQL, HTML/CSS, JavaScript, LaTeX

Developer Tools/Technologies/Frameworks: VS Code, Eclipse, Linux, Git, GitHub, MariaDB, Jupyter Notebook

Certificates: Amazon Web Services Certified Solutions Architect Associate

Leadership / Extracurricular

MHacks 2025

Participant

September 2025

University of Michigan

- Designed and developed Spanish-language learning application using Typescript/HTML frontend and Python backend
- Implemented tokenization/dictionary functionality for easy vocabulary learning, used 2-model system consisting of summarizer model and chatbot model to reduce token usage and store facts from previous conversations